

```
// Automatic Irrigation System

const int sensorPin = A0;    // Soil Moisture Sensor connected to A0

const int relayPin = 7;      // Relay Module connected to Digital Pin 7


int moistureValue;

int threshold = 500;         // Adjust according to your soil condition


void setup()

{

    Serial.begin(9600);

    pinMode(relayPin, OUTPUT);

    // Initially turn OFF pump
    digitalWrite(relayPin, HIGH);
}

void loop()

{

    moistureValue = analogRead(sensorPin);

    Serial.print("Moisture Value: ");
    Serial.println(moistureValue);

    // Dry Soil
    if (moistureValue > threshold)
    {

        Serial.println("Soil is Dry");
        Serial.println("Pump ON");

        digitalWrite(relayPin, LOW); // Turn ON relay
    }

    // Wet Soil
    else
    {

        Serial.println("Soil is Moist");
        Serial.println("Pump OFF");

        digitalWrite(relayPin, HIGH); // Turn OFF relay
    }
}
```

```
delay(2000);
```

```
}
```